

TECHNICAL PROJECT REPORT

AI-Powered Job Recruitment & Candidate Management System

Built on Salesforce CRM with Agentforce AI Framework

Platform Environment: Salesforce Developer Edition Org (`orgfarm-6a0a73fb2f-dev-ed`)

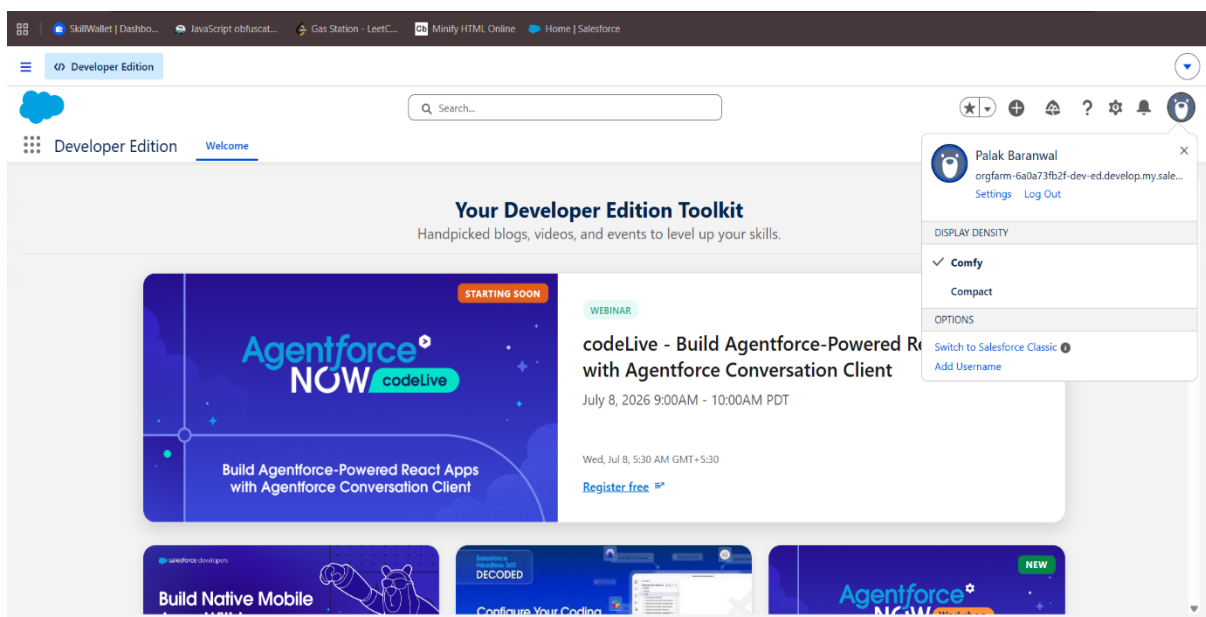


Figure 1.1: Salesforce Developer Edition active org validation for Palak Baranwal.

System Architect / Administrator: Palak Baranwal

Team ID: SWTID-2026-2645

Skill Wallet ID: SWUID2026234516

Email ID: palakbaranwal05@gmail.com

Academic / Internship Track:

SmartBridge Virtual Internship Program

Salesforce Certified Administrator with AI Agentforce Specialization

Date of Submission: July 15, 2026

Certificate of Completion

To Whom It May Concern

This is to certify that the project titled "**AI-Powered Job Recruitment & Candidate Management System**" has been successfully configured and executed by **Palak Baranwal** under the SmartBridge Virtual Internship Program - Salesforce Certified Administrator with AI Agentforce Specialization.

The system metadata, data schemas, automated record-triggered processing frameworks, validation rules, and generative AI models documented within this technical report represent a fully functional, deployment-ready platform solution. All development work detailed herein was conducted natively within a zero-custom-Apex Salesforce configuration layer, adhering directly to enterprise standards for platform cloud scalability, data integrity governance, and role-based access controls.

SmartBridge Program Coordinator

Technical Mentorship Directorate

Acknowledgement

I would like to express my sincere technical gratitude to the core engineering mentors and evaluation coordinators of the SmartBridge Virtual Internship Program. The structured learning tracks covering advanced Salesforce administration, declarative process optimization via Flow Builder, and enterprise generative AI deployment via the Agentforce framework provided the professional foundations required to configure this system architecture.

I also extend acknowledgment to the global Salesforce Trailhead developer network for providing reference models concerning relational data protection, cross-object field mapping, and secure profile validation parameters. Finally, I thank my peer teams for their collaborative feedback during the programmatic testing and simulation loops of this system build.

Table of Contents

1. Executive Summary & Business Rationale
2. Project Scope & Functional Lifecycle
3. Relational Data Model & Schema Architecture
4. Custom Fields & Relationships Directory
5. User Interface Customization & Workspace Design
6. Data Integrity Architecture & System Validation Rules
7. Deduplication & Matching Governance Frameworks
8. Documented Messaging Architecture (Templates & Alerts)
9. Declarative Automation Engine (Flow Builder Architecture)
10. Enterprise Approval Routing & Policy Governance
11. Agentforce AI Integration & Trust Guardrails
12. Security Architecture, Profiles, & Access Control
13. Data Migration & Legacy Ingestion Strategy
14. System Verification & Quality Testing Logs
15. Change Management, Support, & Platform Maintenance
16. Strategic Platform Advantages & Future Scope
17. Conclusion & Reference Catalog

1. Executive Summary & Business Rationale

Modern enterprise organizations process hundreds of candidate applications daily during scaling periods. Traditional talent acquisition workflows that rely heavily on manual tracking systems, standalone spreadsheets, and unlinked email strings create severe operational challenges, including:

- **Data Redundancy:** Inability to trace duplicate candidate records across concurrent recruitment cycles.
- **Application Attrition:** High risk of dropping qualified applicant profiles due to manual routing flaws.
- **Recruitment Latency:** Delayed interview scheduling caused by unstructured coordination loops.
- **Subjective Inconsistency:** Screenings lack standardized evaluation criteria, introducing bias.
- **Operational Opacity:** Management tiers lack visibility into live applicant pipelines and aggregate team productivity.

To eliminate these operational cracks, this project establishes a centralized, Salesforce CRM-driven recruitment application. This system integrates tracking structures, automates sequential workflows, executes real-time AI-based evaluations, accelerates managerial approval chains, and optimizes recruiter velocity without introducing custom programmatic overhead.

2. Project Scope & Functional Lifecycle

The implementation scope covers the deployment of a scalable recruitment platform built exclusively via declarative tools, advanced automations, and Agentforce AI within the Salesforce Lightning ecosystem. Programmatic components (Apex, Visualforce, LWC) are omitted to minimize technical debt and ensure long-term ease of maintenance. The application drives and tracks the complete, end-to-end recruitment lifecycle through five sequential operational phases:

Candidate → Screening → Interview → Offer → Hiring

Core development steps are split across five distinct engineering phases: Requirement Analysis and Planning; Backend Schema Configuration; UI/UX Customization; Data Migration & Deduplication Security; and Deployment Validation with Change Management.

3. Relational Data Model & Schema Architecture

The structural data layer is constructed using highly optimized, custom relational schemas mapped directly to native Salesforce object structures. The architecture focuses on two core custom entities built within the Object Manager layer:

- **Job Opening (Job_Opening__c):** Represents a single open job position. It tracks corporate structural metrics such as department parameters, target vacancy counts, job status tracking picklists, and baseline financial allocations. The record name field is configured as **Job Title** (Data Type: Text).

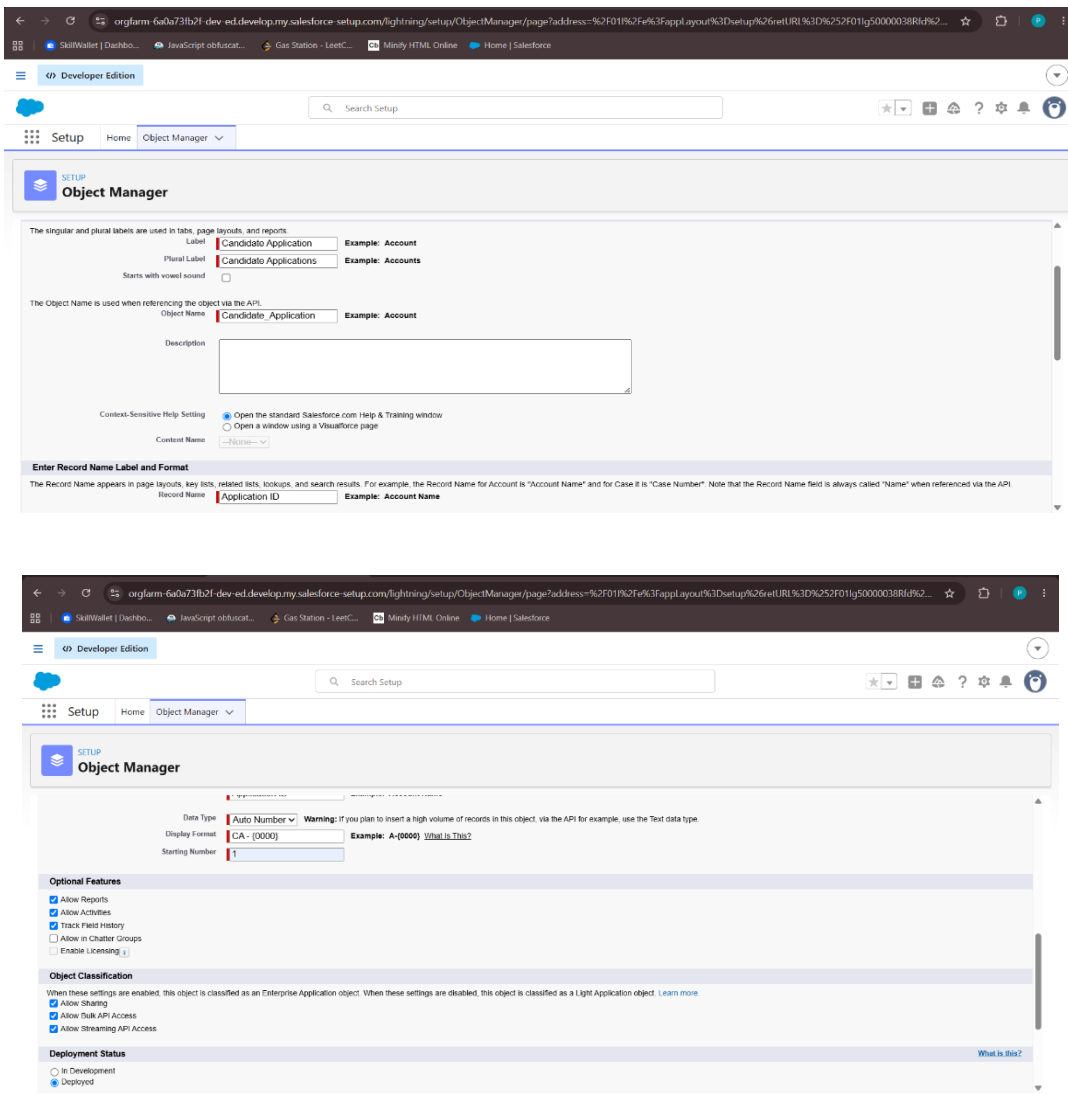
The figure consists of two screenshots from the Salesforce Setup interface, specifically the 'New Custom Object' page for 'Job Opening'.

The top screenshot shows the 'Custom Object Definition Edit' section. The 'Custom Object Information' tab is active. The 'Label' is 'Job Opening' (Example: Account), and the 'Plural Label' is 'Job Openings' (Example: Accounts). The 'Object Name' is 'Job_Opening' (Example: Account). The 'Description' field is empty. The 'Context Sensitive Help Setting' is set to 'Open the standard Salesforce.com Help & Training window'. The 'Context Name' is 'None'.

The bottom screenshot shows the 'Record Name' field set to 'Job Title' (Example: Account Name). The 'Data Type' is 'Text'. A warning message states: 'Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.' The 'Optional Features' section includes 'Allow Reports' (checked), 'Allow Activities' (checked), 'Track Field History' (checked), 'Allow in Chatter Groups' (unchecked), and 'Enable Licensing' (unchecked). The 'Object Classification' section includes 'Allow Sharing' (checked), 'Allow Bulk API Access' (checked), and 'Allow Streaming API Access' (checked). The 'Deployment Status' is set to 'Deployed'.

“Figure 3.1: Step-by-step creation of Custom Object Job Opening with history tracking enabled.”

- **Candidate Application (Candidate_Application__c):** A custom junction object configured to manage the recruitment lifecycle. To prevent orphaned transactional entries, the record name field uses an automated sequencing format set to **Application ID** (Data Type: Auto Number, format: CA-{0000}, starting with number 1).



“Figure 3.2: Custom junction object Candidate Application setup with Auto-Number formatting.”

To ensure high relational integrity, the data model maps parent-child object connections using dual ****Master-Detail Relationships**** from the `Candidate_Application__c` child record. It links directly up to the `Job_Opening__c` table and the standard `Contact` (Candidate) object ledger. If a master candidate profile or parent job opening record is deleted, the platform cascade-deletes related candidate applications automatically, keeping data clean and avoiding broken references.

4. Custom Fields & Relationships Directory

The tables below outline the structural custom field allocations configured across objects within the Object Manager layer of the developer environment:

4.1 Standard Contact Object (Candidate Ledger Extensions)

Field Label	Field API Name	Data Type	Configuration Parameters / Picklist Values
Candidate ID	Candidate_ID__c	Auto Number	Display Format: C-{0000} (Starting Number: 1)
Experience Level	Experience_Level__c	Picklist	Fresher, Mid-Level, Senior
Application Status	Application_Status__c	Picklist	New, Screening, Interview, Offer, Hired

English Grammar: L...Data Types in C++...Top Interview 150...SkillWallet | Dashbo...Email - Google SearchFutureSkills PrimeSalesforce CRM Soft...Trailhead | The fun...

Developer Edition

Search Setup

SetupHomeObject Manager

SETUP > OBJECT MANAGER

Contact

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Hierarchy Columns

Scoping Rules

Field LabelCandidate ID

Field NameCandidate_ID

Description

Help Text

Data OwnerUser

Field Usage--None--

Data Sensitivity Level--None--

Compliance Categorization

AvailablePIIHIPAAGDPRPCI

Chosen

General Options

External ID☐ Set this field as the unique record identifier from an external system

Auto Number Options

Auto Number Display FormatC-{0000}

Change Field TypeSaveCancel

English Grammar: L...Data Types in C++...Top Interview 150...SkillWallet | Dashbo...email - Google SearchFutureSkills PrimeSalesforce CRM Soft...Trailhead | The fun...

Developer Edition

SetupHomeObject Manager

Search Setup

Setup > OBJECT MANAGER

Contact

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Hierarchy Columns

Scoping Rules

Contact

New Custom Field

Step 2. Enter the details

Field LabelExperience Level

Values

- Use global picklist value set
- Enter values, with each value separated by a new line
 - Fresher
 - Mid-Level
 - Senior

☐ Display values alphabetically, not in the order entered

☐ Use first value as default value

☒ Restrict picklist to the values defined in the value set

Field NameExperience_Level

Description

PreviousNextCancel

Step 2 of 4

Help for this Page

Einstein

English Grammar: L...Data Types in C++...Top Interview 150...SkillWallet | Dashbo...email - Google SearchFutureSkills PrimeSalesforce CRM Soft...Trailhead | The fun...

Developer Edition

SetupHomeObject Manager

Search Setup

Setup > OBJECT MANAGER

Contact

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Hierarchy Columns

Scoping Rules

Edit Contact Custom Field

Application Status

Custom Field Definition Edit

Field Information

Field LabelApplication Status

Field NameApplication_Status

Description

Help Text

Data OwnerUser

Field Usage--None--

Data Sensitivity Level--None--

Compliance Categorization

- Available
 - PII
 - HIPAA
 - GDPR
 - PCI
- Chosen

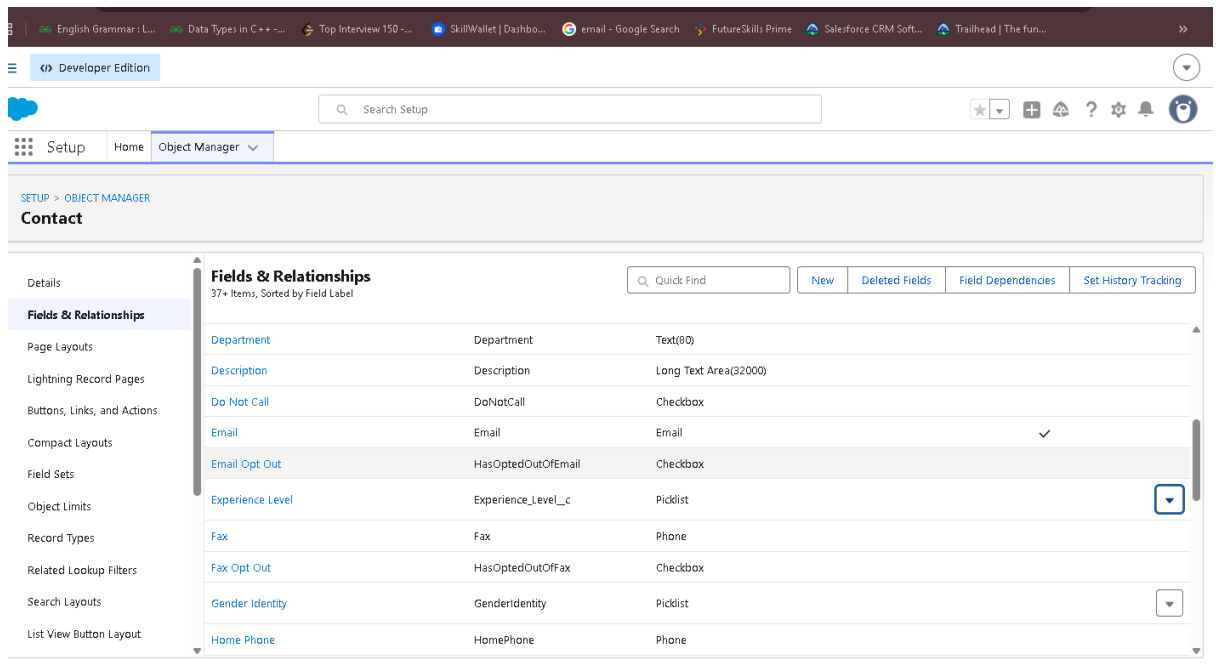
General Options

Required☐ Always require a value in this field in order to save a record

Change Field TypePromote to Global Value SetSaveCancel

Required Information

Einstein



“Figure 4.1: Custom field parameters and synchronization values on the Contact layer.”

4.2 Custom Job Opening Object (Job_Opening__c)

Field Label	Field API Name	Data Type	Configuration Parameters / Picklist Values
Department	Department__c	Picklist	IT, HR, Finance, Sales
Salary Range	Salary_Range__c	Currency	Standard Currency Formatting (18, 2)
Vacancies	Vacancies__c	Number	Integer Formatting (Scale: 0; negative check active)
Job Status	Job_Status__c	Picklist	Open, Closed, On Hold

The screenshot shows the Salesforce Setup interface, specifically the 'Object Manager' for the 'Job Opening' object. The 'Fields & Relationships' tab is selected, displaying a directory matrix of fields. The left sidebar shows navigation options like Details, Fields & Relationships, Page Layouts, etc. The main area shows a table with columns: FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED. The table lists fields such as Created By, Department, Job Status, Job Title, Last Modified By, Owner, Salary Range, and Vacancies.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Department	Department__c	Picklist		
Job Status	Job_Status__c	Picklist		
Job Title	Name	Text(80)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Salary Range	Salary_Range__c	Currency(18, 0)		
Vacancies	Vacancies__c	Number(18, 0)		

“Figure 4.2: Fields & Relationships directory matrix for Job Opening custom object.”

4.3 Custom Candidate Application Object (Candidate_Application__c)

Field Label	Field API Name	Data Type	Configuration Parameters / Picklist Values
Candidate	Candidate__c	Master-Detail	Parent Target Connection: Contact (Required)
Job Opening	Job_Opening__c	Master-Detail	Parent Target Connection: Job_Opening__c (Required)
Application Date	Application_Date__c	Date	Standard Date Tracking
AI Candidate Score	AI_Candidate_Score__c	Number	Integer Formatting (Enforced Bounds: 0 to 100)
Offer Amount	Offer_Amount__c	Currency	Standard Currency Ledger Layout (Scale: 18, 2)

Field Label	Field API Name	Data Type	Configuration Parameters / Picklist Values
Application Status	Application_Status__c	Picklist	Active, Inactive, Offer Submitted
Approval Status	Approval_Status__c	Picklist	Pending, Approved, Rejected

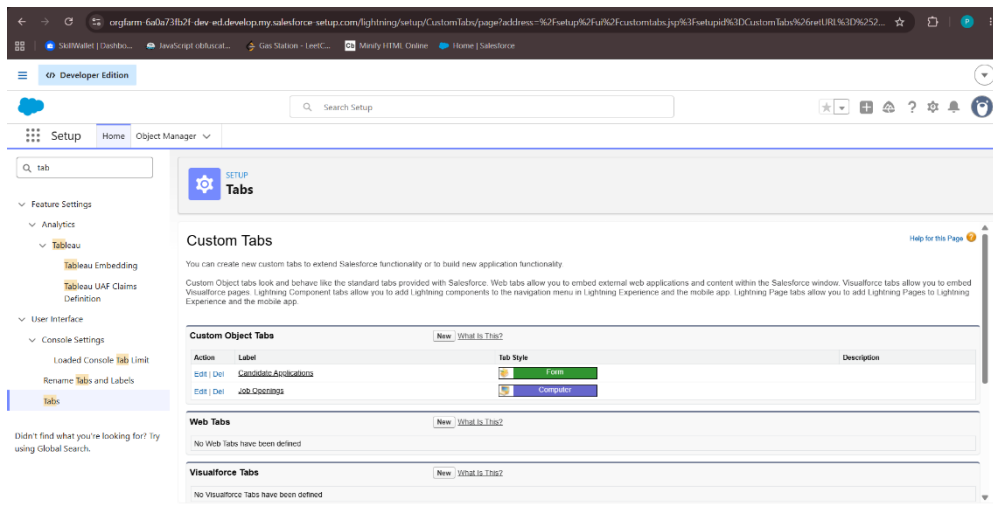
FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
AI Candidate Score	AI_Candidate_Score__c	Number(18, 0)		
Application Date	Application_Date__c	Date		
Application ID	Name	Auto Number		✓
Application Status	Application_Status__c	Picklist		
Approval Status	Approval_Status__c	Picklist		
Candidate	Candidate__c	Master-Detail(Contact)		✓
Created By	CreatedById	Lookup(User)		
Job Opening	Job_Opening__c	Master-Detail(Job Opening)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Offer Amount	Offer_Amount__c	Currency(18, 0)		

“Figure 4.3: Structural database schema directory for Candidate Application junction table.”

5. User Interface Customization & Workspace Design

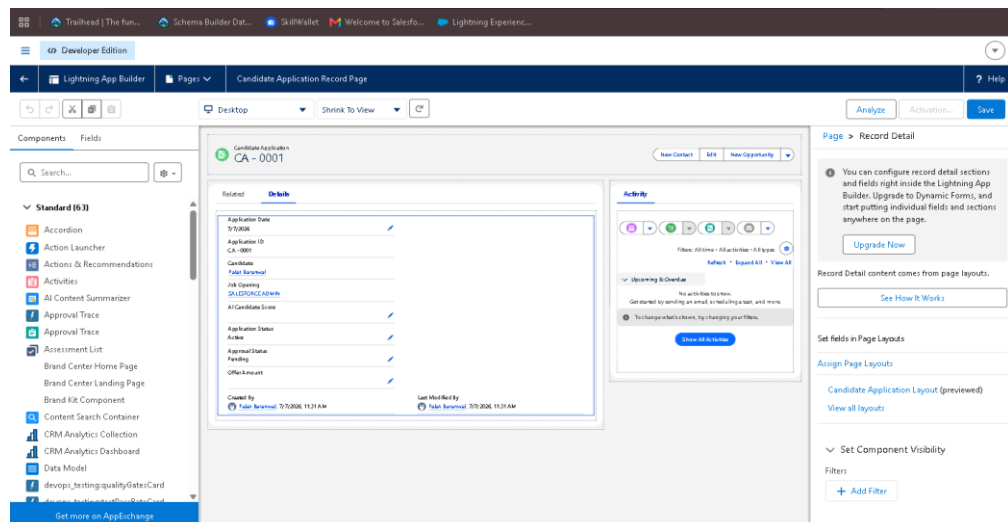
To deliver an intuitive and productivity-driven workspace, user interface components are structured within a custom Lightning console setup:

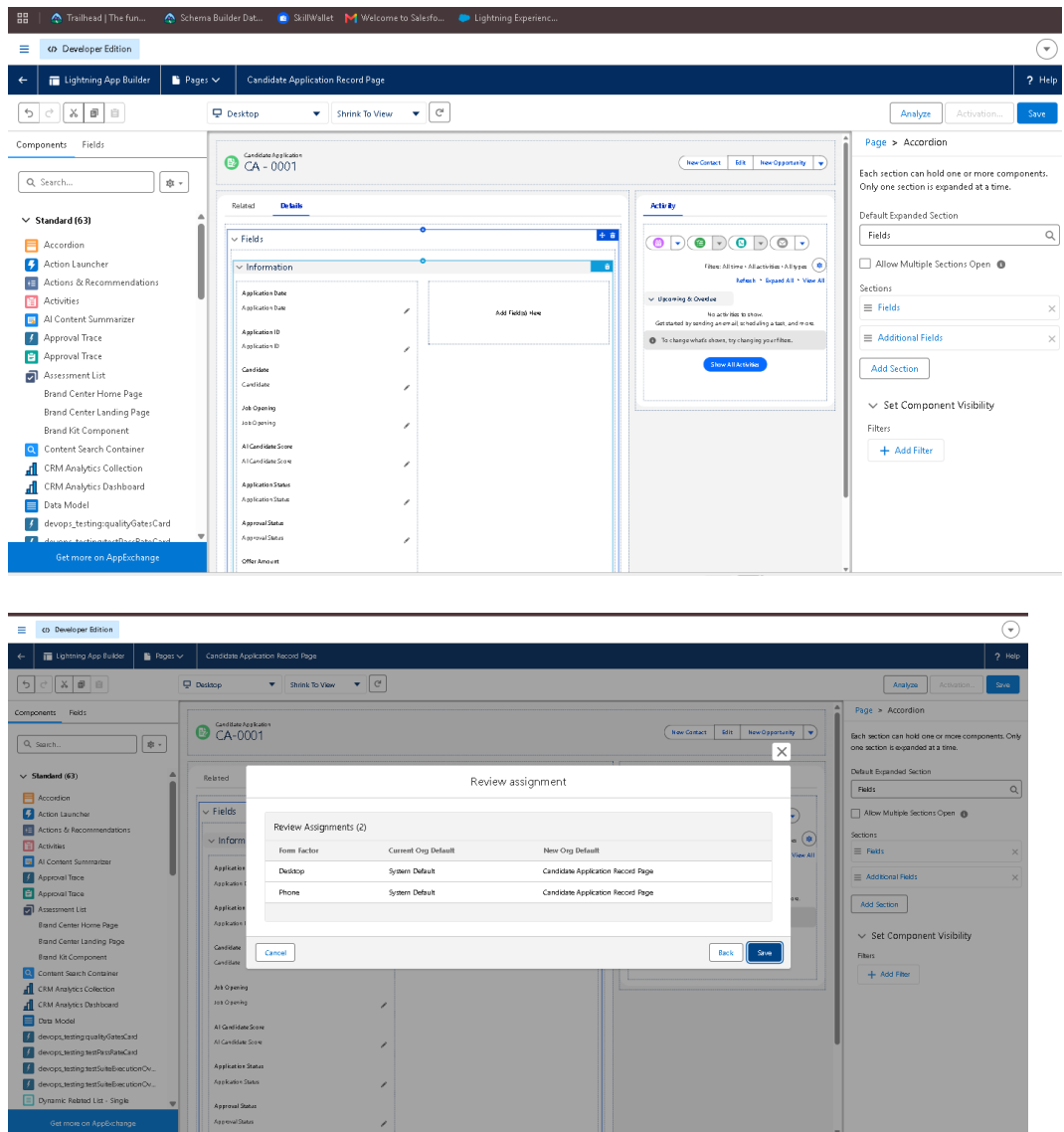
- **Custom Interface Tabs Setup:** Custom object navigation entries are generated to control data access. The **Job Openings** tab uses the `Computer` visual style, and the **Candidate Applications** tab uses the `Form` visual design. Both tabs are set to append automatically to personalized user navigation profiles.



“Figure 5.1: Navigation Custom Object Tabs wizard directory deployment.”

- **Lightning Console Application Configuration:** All structural objects are grouped under a unified Lightning application labeled "**AI-Powered Recruitment & Candidate Management System**". Navigation settings include Candidate Applications, Candidates (Contact), Job Openings, Accounts, Reports, and Dashboards. Access rights are granted exclusively to System Administrator, Recruiter, and Hiring Manager security profiles.
- **Dynamic Forms Architecture Deployment:** Standard record page views are upgraded to **Dynamic Forms** layers. This approach allows administrators to configure field visibility and section arrangements dynamically from the Lightning App Builder without relying on rigid page layouts. This upgrade improves record load performance on both desktop and mobile devices.





“Figure 5.2: Upgrading and activating Lightning Record Pages to cross-platform Dynamic Forms.”

6. Data Integrity Architecture & System Validation Rules

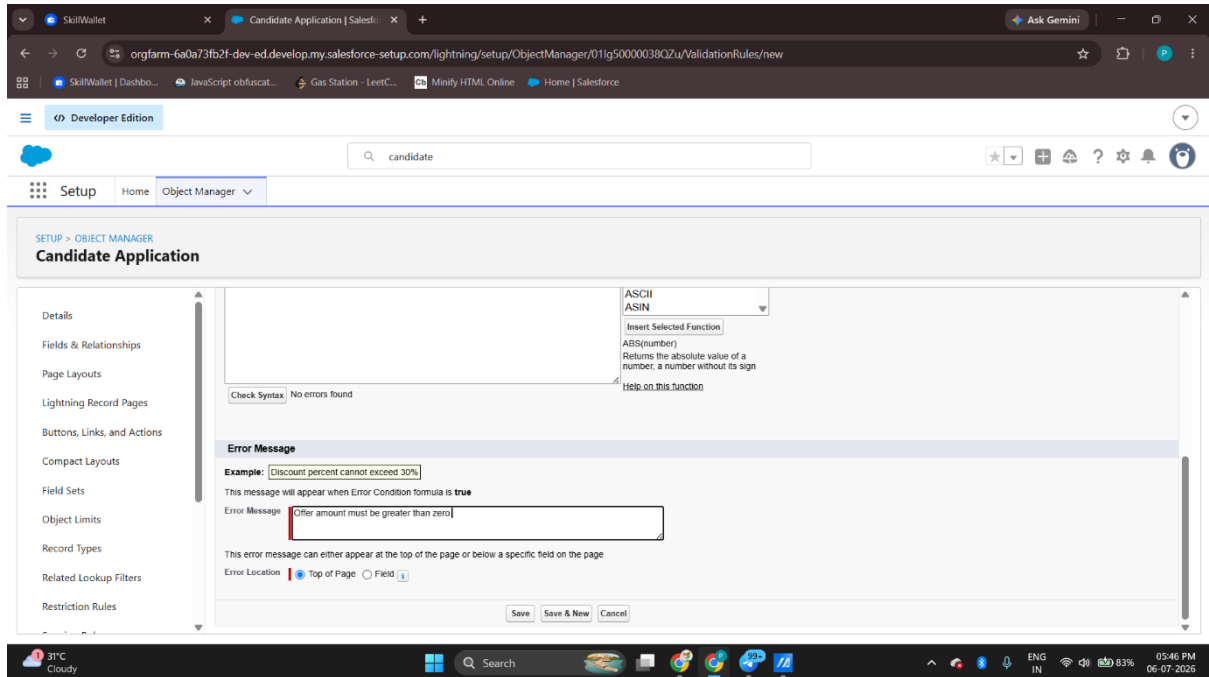
To enforce high data quality and provide clean information for downstream AI workflows, the system uses strict declarative validation formulas. If invalid data values are entered, database saves are blocked, and explicit error labels display to the user.

Rule 1: Offer Amount Validation

- **Target Entity Object:** Candidate Application (Candidate_Application__c)
- **Unique Developer Identifier:** Offer_Amount_Validation
- **Error Condition Evaluation Formula:**

```
Offer_Amount__c <= 0
```

- **System Presentation Message String:** "Offer amount must be greater than zero."
- **Error Layout Mapping:** Top of page window.



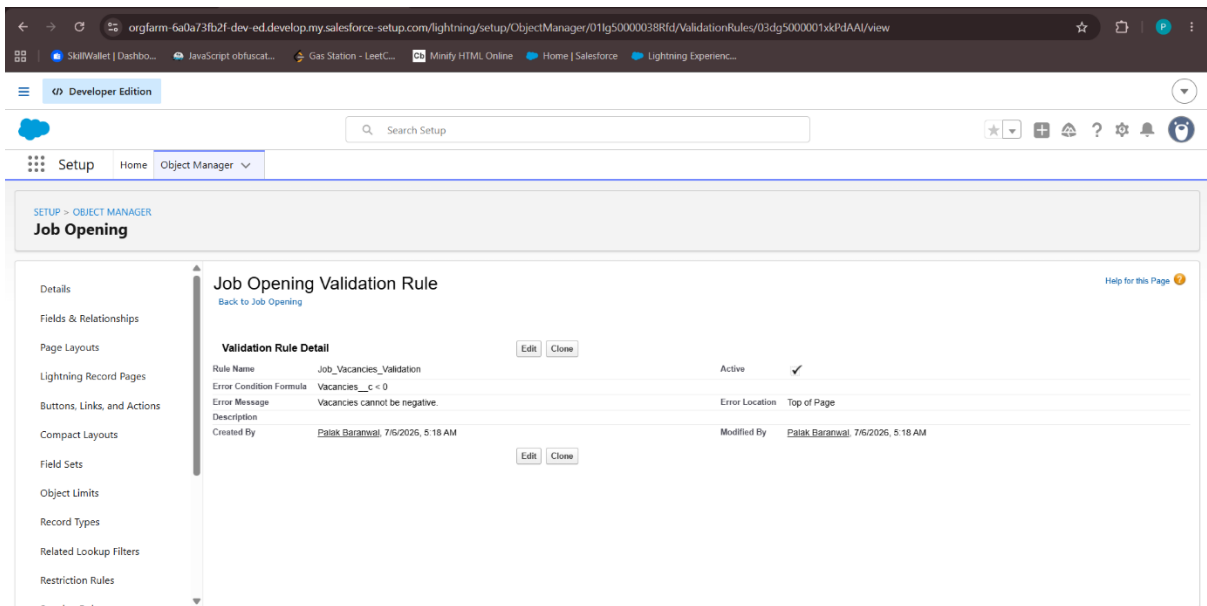
“Figure 6.1: Enforcing positive compensation constraints via Offer Amount validation formula.”

Rule 2: Job Vacancies Validation

- **Target Entity Object:** Job Opening (Job_Opening__c)
- **Unique Developer Identifier:** Job_Vacancies_Validation
- **Error Condition Evaluation Formula:**

Vacancies__c < 0

- **System Presentation Message String:** "Vacancies cannot be negative."
- **Error Layout Mapping:** Top of page window.



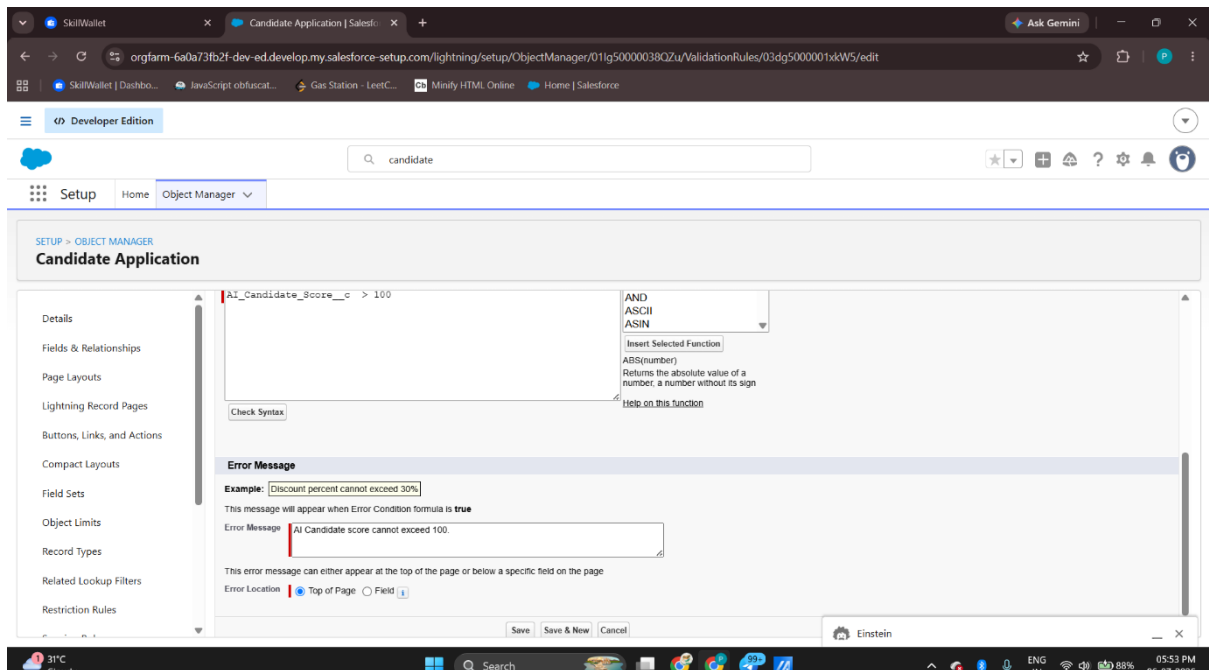
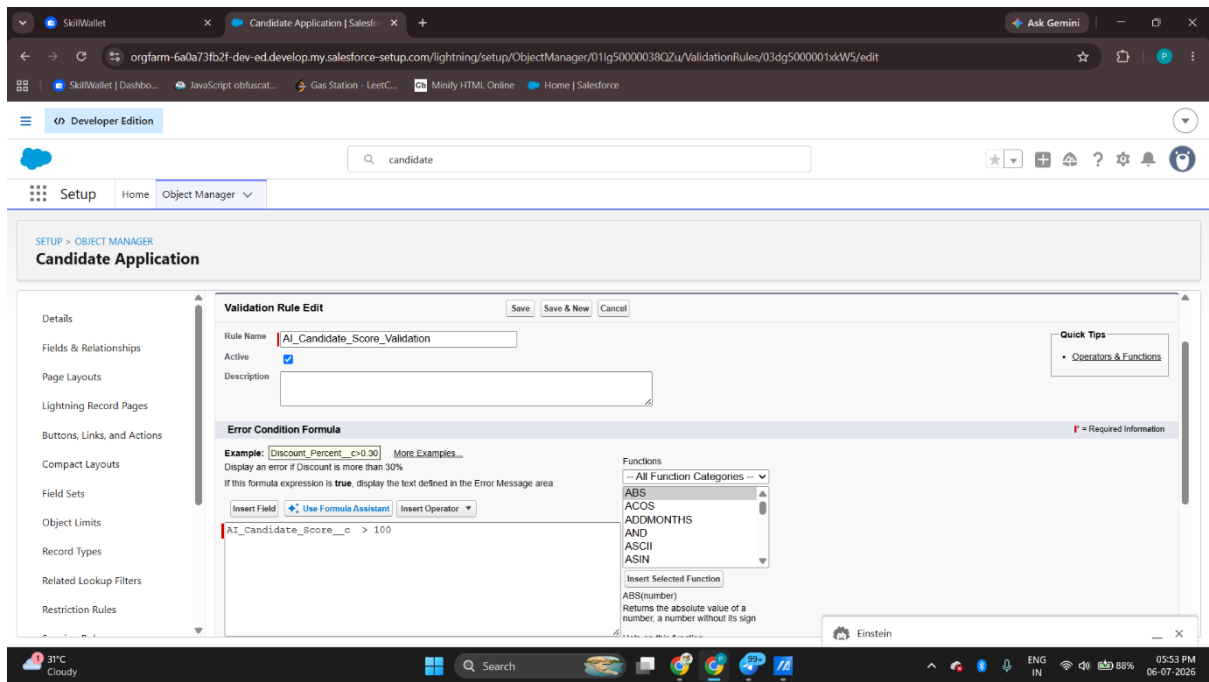
“Figure 6.2: Enforcing vacancy data compliance verified under creator Palak Baranwal.”

Rule 3: AI Candidate Score Validation

- **Target Entity Object:** Candidate Application (Candidate_Application__c)
- **Unique Developer Identifier:** AI_Candidate_Score_Validation
- **Error Condition Evaluation Formula:**

```
AI_Candidate_Score__c > 100
```

- **System Presentation Message String:** "AI Candidate score cannot exceed 100."
- **Error Layout Mapping:** Top of page window.



“Figure 6.3: Integrity validation ensuring score matrices do not breach upper limits.”

7. Deduplication & Matching Governance Frameworks

Duplicate candidate records can fragment history tracking and skew downstream AI evaluation scores. The system resolves this vulnerability by establishing a native deduplication engine at the Candidate (Contact) object tier:

19. **Custom Matching Rule (Unique Candidate - Email & Phone):** This rule uses a multi-field matching matrix to identify duplicate records. It scans newly submitted candidate entries using **Exact Matching Methods** on the `Email` and `Phone` fields. The **Match Blank Fields** parameter is enabled to catch matching profiles across partial data entries.
20. **Custom Duplicate Rule (Unique Candidate Email and Phone):** This rule enforces system policies when matching entries are detected. It triggers on both **Record Creation** and **Record Modification**, executing **Allow and Report** behavior. This allows data tracking while logging duplicate attempts for administrative review. When triggered, it presents a clear alert string to the user: "Student Email and Phone Number must be unique."

8. Documented Messaging Architecture (Templates & Alerts)

Automated communication loops use standard Classic plain-text email templates to ensure high delivery rates across corporate firewalls:

8.1 Configuration Template:

High_Priority_Candidate_Application_Notification`

Subject Header String: High-Priority Candidate Application Assigned to You

```
Hello {!Candidate_Application__c.OwnerFullName}, A candidate application has been identified as high priority by AI and requires immediate attention. Candidate Application Details: Candidate Name: {!Candidate_Application__c.Candidate__r.Name} Job Opening: {!Candidate_Application__c.Job_Opening__r.Name} Experience Level: {!Candidate_Application__c.Candidate__r.Experience_Level__c} AI Candidate Score: {!Candidate_Application__c.AI_Candidate_Score__c} Application Date: {!Candidate_Application__c.Application_Date__c} Please contact the candidate and proceed with the recruitment process within 24 hours. Thank you, AI Recruitment Management System
```

8.2 Configuration Template:

High_Salary_Candidate_Offer_Approval_Notification`

Subject Header String: Candidate Offer Awaiting Salary Approval

```
Dear {!Candidate_Application__c.OwnerFullName}, A candidate job offer with a high salary has been submitted for your approval. Please review the candidate application details below: Candidate Offer Details: Candidate Name: {!Candidate_Application__c.Candidate__r.Name} Job Opening: {!Candidate_Application__c.Job_Opening__r.Name} Experience Level: {!Candidate_Application__c.Candidate__r.Experience_Level__c} Proposed Offer Amount:
```

```
{!Candidate_Application__c.Offer_Amount__c} AI Candidate Score:
{!Candidate_Application__c.AI_Candidate_Score__c} Application
Status: {!Candidate_Application__c.Application_Status__c} Please
take the necessary action at your earliest convenience. Thank
you, AI Recruitment Management System
```

8.3 Configuration Template:

`Candidate\_Application\_Approval\_Status\_Notification`

Subject Header String: Candidate Offer Approval Decision

```
Hello {!Candidate_Application__c.OwnerFullName}, The approval
process for a candidate offer has been completed. Application
Summary: Candidate Name:
{!Candidate_Application__c.Candidate__r.Name} Job Opening:
{!Candidate_Application__c.Job_Opening__r.Name} Approval Status:
{!Candidate_Application__c.Approval_Status__c} Final Offer
Amount: {!Candidate_Application__c.Offer_Amount__c} Please
proceed with the next steps in the recruitment process based on
the approval outcome. Thank you, AI Recruitment Management System
```

8.4 System Email Alerts Execution Mapping

- **Alert 1 (High Priority Ingestion Notification):** Built on Candidate_Application__c. Tied to template 8.1. Routes automatically to the target user field: Candidate Application Owner.
- **Alert 2 (High Salary Offer Submitted Alert):** Built on Candidate_Application__c. Tied to template 8.2. Routes automatically to the target user field: Manager.
- **Alert 3 (Approval Sequence Outcome Status Alert):** Built on Candidate_Application__c. Tied to template 8.3. Routes automatically to the target user field: Candidate Application Owner.

The screenshot displays the Salesforce 'Email Alerts' configuration page. The top navigation bar includes 'Developer Edition' and a search bar. The left sidebar shows 'Setup', 'Home', and 'Object Manager' tabs, with 'Email Alerts' selected under 'Process Automation'. The main content area is titled 'Email Alerts' and contains a table of configured alerts.

Action	Description *	Email Template Name	Object	Last Modified Date
Edit Del	Candidate Application Approval Status Email Alert	Candidate Application Approval Status Notification	Candidate Application	7/6/2026
Edit Del	High Priority Candidate Application Email Alert	High Priority Candidate Application Notification	Candidate Application	7/6/2026
Edit Del	High Salary Candidate Offer Approval Email Alert	High Salary Candidate Offer Approval Notification	Candidate Application	7/6/2026

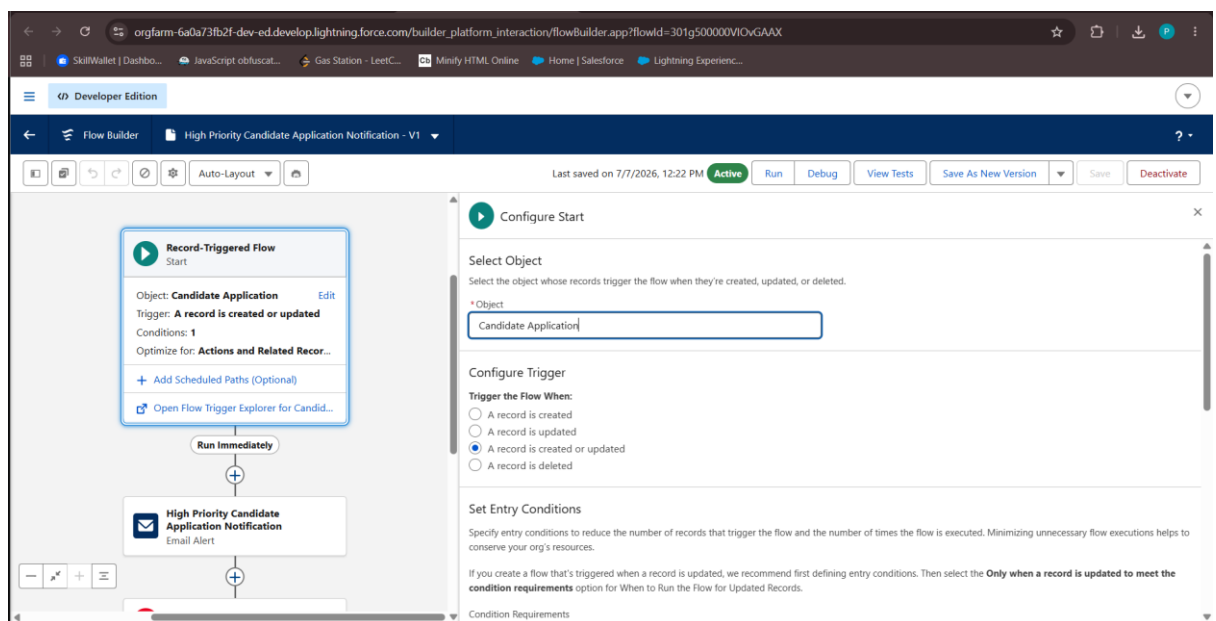
“Figure 8.1: Unified automated system Email Alerts registry catalog.”

9. Declarative Automation Engine (Flow Builder Architecture)

The system replaces manual milestone tracking with four declarative business process automation paths configured in Flow Builder:

Flow 1: High Priority Candidate Application Notification (Record-Triggered)

Triggers when a `Candidate_Application__c` record is created or updated. It filters operations using the entry criteria formula: `AI_Candidate_Score__c >= 85`. The flow is set to execute **every time a record is modified to meet the condition requirements**, optimized for ***Actions and Related Records***. When triggered, it calls the `High Priority Candidate Application Email Alert` action node, mapping the current record identity token (`{!$Record.Id}`) to pass application details dynamically.

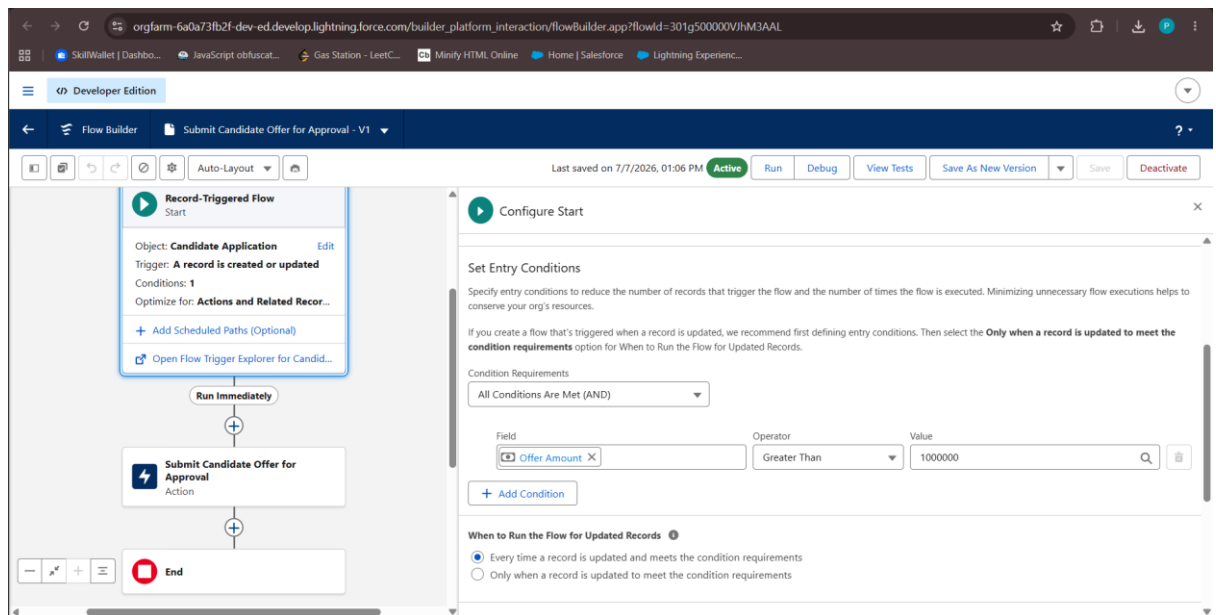


“Figure 9.1: Building and syntax-checking the record-triggered high priority notification flow.”

Flow 2: High Salary Offer Submission for Approval (Record-Triggered)

Triggers when a `Candidate_Application__c` record is created or updated. It filters operations using the entry criteria formula: `Offer_Amount__c > 1000000`. Optimized for ***Actions and Related Records***, it triggers a core platform action node labeled ***Submit for Approval***. This action targets the `High Salary Candidate Offer`

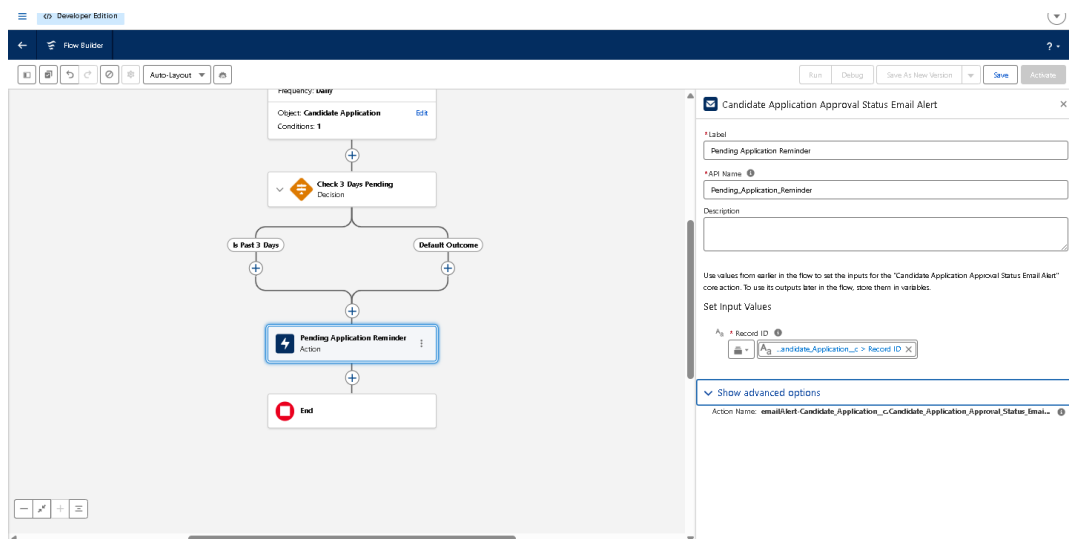
Approval` process template, automatically locking the record data row from further changes and routing it to the designated reviewer.

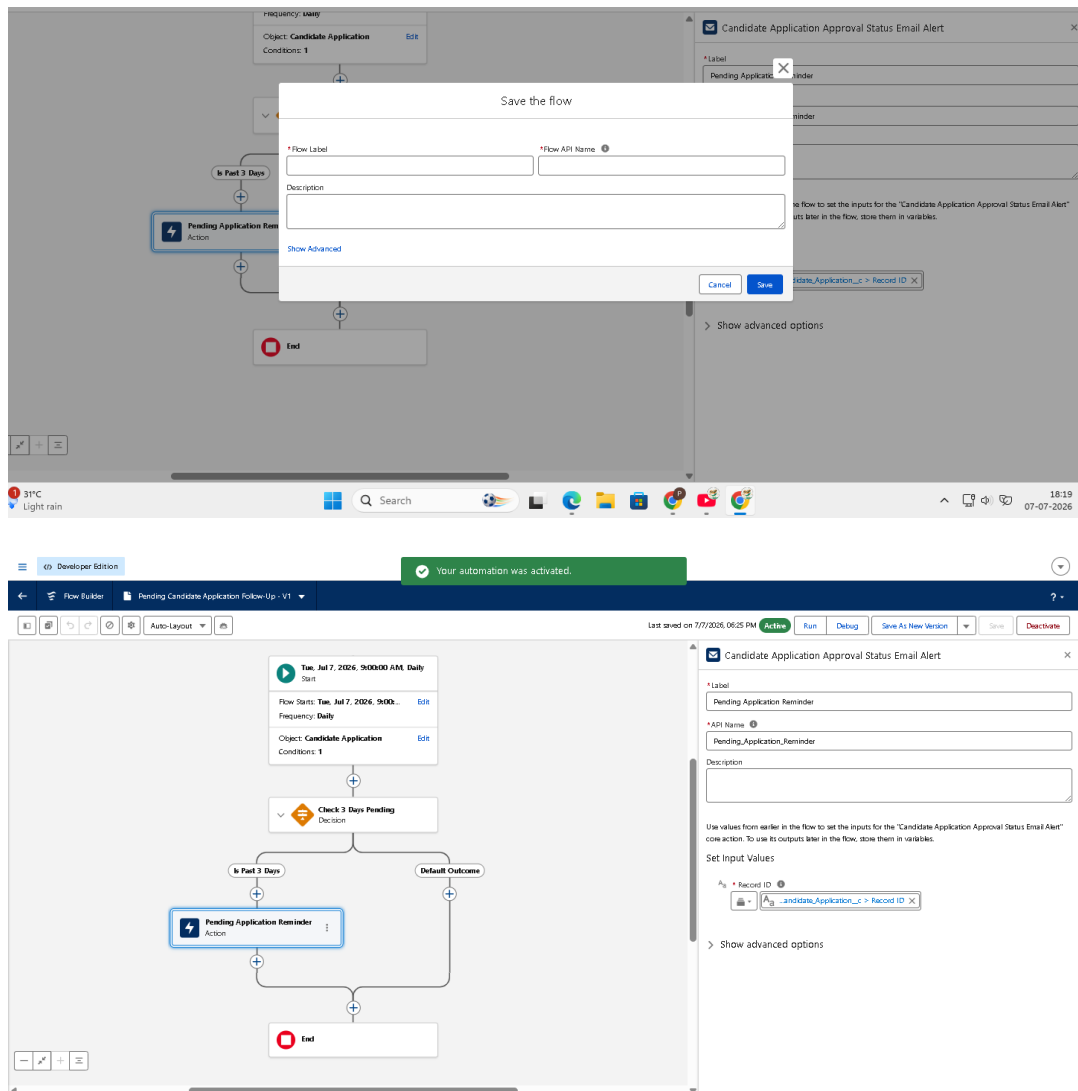


“Figure 9.2: Activated operational canvas for the automated approval submission workflow flow.”

Flow 3: Pending Candidate Application Follow-Up (Scheduled-Triggered)

Runs on a automated daily schedule at a designated morning interval (e.g., 9:00 AM) to evaluate all candidate records against two criteria conditions: `Approval_Status__c == 'Pending' AND LastModifiedDate < TODAY() - 3`. When records match this criteria loop, the flow calls an email alert action node labeled ****Pending Application Reminder****. This system automatically sends follow-up notifications to the reviewer, preventing candidates from getting stuck in the pipeline.

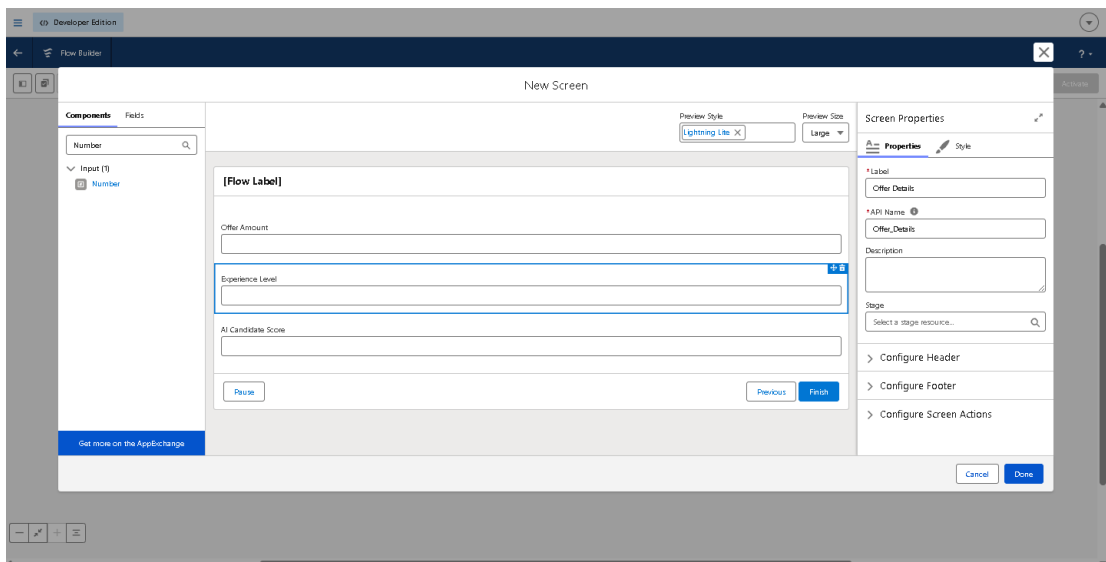
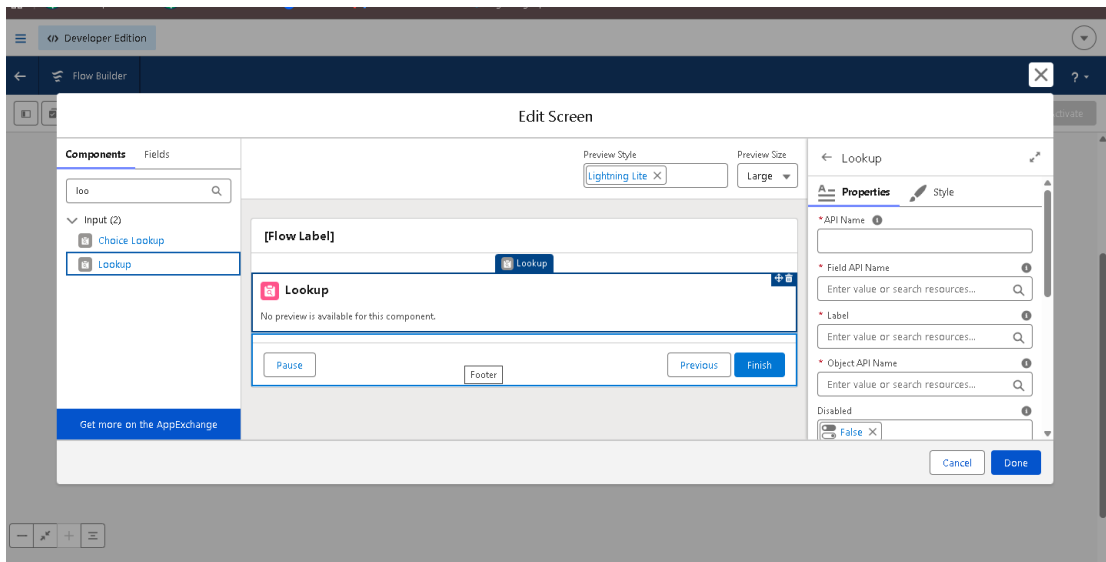




“Figure 9.3: Live production-ready scheduled batch automation engine deployment canvas.

Flow 4: Candidate Application Creation Wizard (Interactive Screen Flow)

Provides recruiters with a guided multi-screen interface to register applications uniformly. **Screen 1 (Candidate Details)** captures parent lookup selections for Candidate and Job Opening along with the Application Date. **Screen 2 (Offer Details)** captures manual inputs for proposed Offer Amount, Experience Level, and baseline AI Candidate Score fields. A background assignment calculation step validates financial parameters before routing variables to a `Create Records` node to insert the target `Candidate_Application_c` row. Upon successful insertion, **Screen 3 (Confirmation)** displays validation text: "Candidate Application Created Successfully". (API Identifier: `Candidate_Application_Screen_Flow`).



“Figure 9.4: Final interactive UI fields layout block mapping inside Screen Flow wizard component.”

10. Enterprise Approval Routing & Policy Governance

To protect institutional budgets, the system locks high-value compensation offers until they are reviewed and approved by an authorized manager. This structural governance track is handled by an active, multi-action Classic Approval Process definition:

Process Schema Blueprint: `High Salary Candidate Offer Approval`

- **Target Base Object Binding:** Candidate Application (Candidate_Application__c)
- **Process Entry Filtering Logic Matrix:**

```
(Candidate_Application__c.Offer_Amount__c > 1000000) AND  
(Candidate_Application__c.Application_Status__c == 'Offer  
Submitted')
```

The screenshot shows the Salesforce Setup interface. A 'Lookup' dialog is open, allowing selection of picklist values. Below it, the 'Specify Entry Criteria' section is visible, where criteria for the approval process are defined. The criteria table is as follows:

Field	Operator	Value
Candidate Application: Offer Amount	greater than	1000000
Candidate Application: Application Status	equals	Offer Submitted
--None--	--None--	
--None--	--None--	
--None--	--None--	

Below the table, there is an error message: "Error: Picklist value 'Offer Submitted' does not exist for 'Application Status' field." The interface also includes navigation buttons like 'Previous', 'Next', and 'Cancel'.

“Figure 10.1: Defining defensive budgetary entry thresholds for compensation approval.”

- **Record Data Lock Policy:** Records are locked during evaluation; only System Administrators retain manual edit permissions.
- **Approver Routing Rules:** Automatically routes records based on the organization's user relationship schema, targeting the ****Manager**** listed on the submitter's User record.

The screenshot shows the Salesforce Setup interface for 'Approval Processes'. The 'New Approval Step' configuration is displayed, showing 'Step 3: Select Assigned Approver'. The configuration options are as follows:

- ☐ Let the submitter choose the approver manually.
- ☒ Automatically assign using the user field selected earlier: (Manager)
- ☐ Automatically assign to approver(s).
- ☐ The approver's delegate may also approve this request. [1]

The interface includes navigation buttons like 'Previous', 'Save', and 'Cancel'.

“Figure 10.2: Enforcing organizational matrix reporting directly to the owner's Manager.

- **Assigned Template Mapping:** High Salary Candidate Offer Approval Notification
- **Approval Page Layout Display:** Candidate Name, Job Opening, Offer Amount, AI Candidate Score, Application Status.

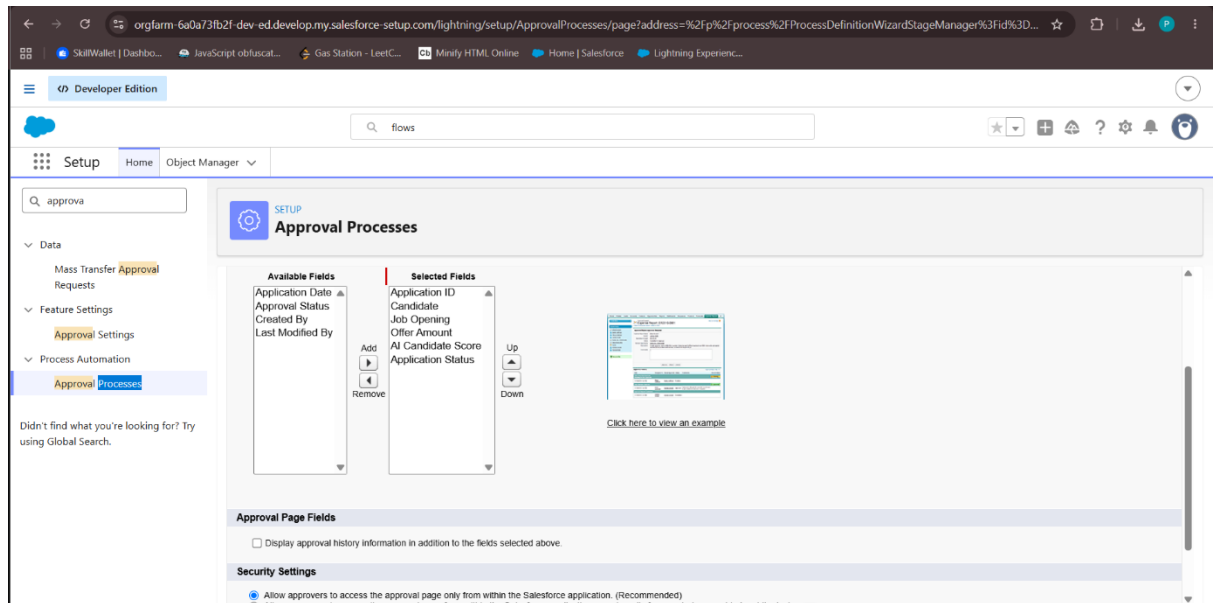


Figure 10.3: Layout properties configuration for the manager approval presentation pane.”

- **Authorized Submitters Ledger:** Restructured to target the ****Record Owner****, allowing frontline recruiters to launch approval routing directly from the interface layout.

Configured Sequence Process Actions

31. **Initial Submission Actions:** Locks record access and fires Alert 2 to notify the manager.
32. **Final Approval Step Actions (Field Update):** Triggers automatically upon managerial sign-off. Runs a field update that changes the `Approval_Status__c` picklist value to **Approved**.
33. **Final Rejection Step Actions (Field Update):** Triggers if a manager denies the compensation plan. Runs a field update that changes the `Approval_Status__c` picklist value to **Rejected**.

Developer Edition

Setup Home Object Manager

Search Setup

Field Updates

Define the field update, including the object associated with the workflow rule, approval process, or entitlement process, the field to update, and the value to apply. Note that the field to update may be on a related object. Fields are shown only for the type that you select.

Field Update Edit Save Save & New Cancel

Identification Required Information

Name SetApproval Status to P
 Unique Name SetApproval_Status_N
 Description
 Object Candidate Application
 Field to Update Approval Status
 Field Data Type Picklist
 Re-evaluate Workflow Rules after Field Change ☐

Specify New Field Value

Picklist Options

☐ The value above the current one
☐ The value below the current one
☒ A specific value Rejected

Save Save & New Cancel

“Figure 10.4: Constructing automated Field Updates to lock status changes upon rejection.”

- Select All Records Should enter this Step

Approval Processes

New Approval Step Help for this Page

Step 2. Specify Step Criteria Step 2 of 3

Specify whether a record must meet certain criteria before entering this approval step. If these criteria are not met, the approval process can skip to the next step, if one exists. [Learn more](#)

Specify Step Criteria

☒ All records should enter this step
☐ Enter this step if the following criteria are met -else approve record

Previous Next Cancel

Step 10: Activate the Approval Process

- Click View Approval Process Detail Page
- Click Activate (top-right)
- List

“Figure 10.5: Executing full process path structure deployment and final activation loops.”

11. Agentforce AI Integration & Trust Guardrails

Rather than conversational chat windows, the system integrates generative AI models directly into standard record pages via Agentforce and Prompt Builder templates. This provides data insights directly to hiring managers within their daily workspace.

11.1 System Trust Guardrails & Governance Controls

To prevent data leaks and protect candidate privacy during external large language model (LLM) evaluations, the platform uses native **AI Trust & Permissions** controls:

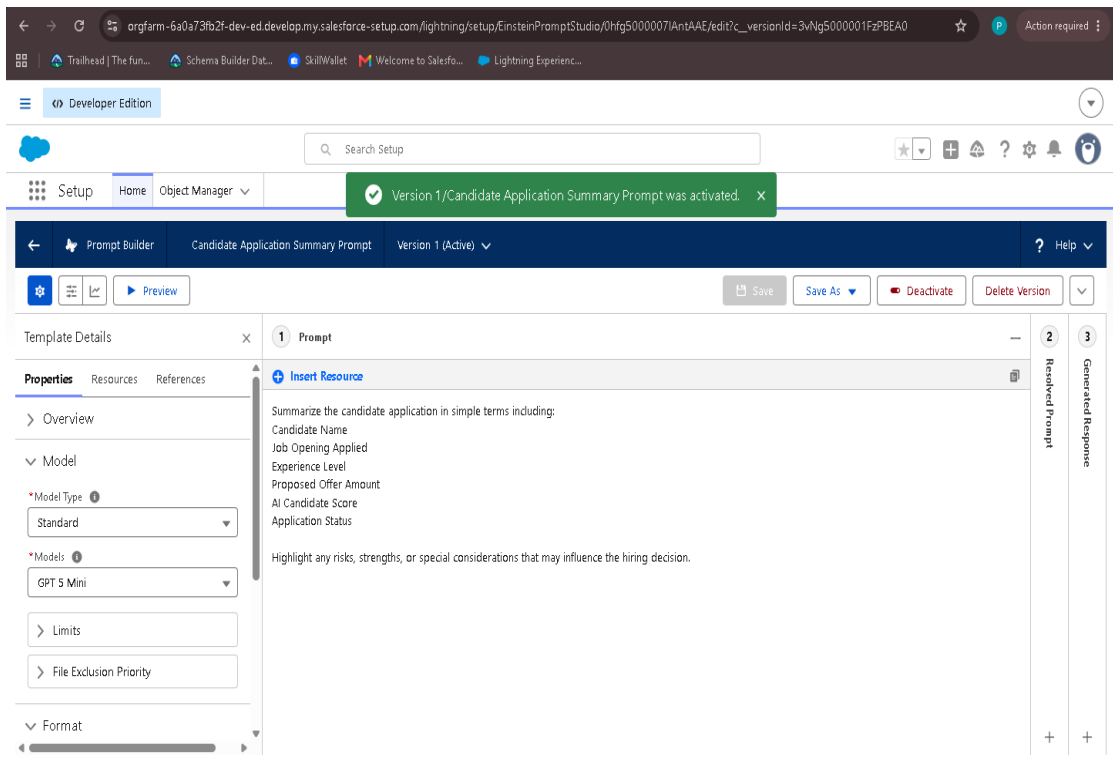
- **Data Masking:** Automatically scrambles personal candidate communication parameters before information passes out of the organizational boundary.
- **Zero Data Retention Policy:** Enforces rules ensuring no prompt strings or context logs are cached or used for external model training.
- **System Audit Logging:** Tracks all prompt template runs and AI generations for administrative security review.
- **Access Profiles:** Restricted to authorized System Administrator and Hiring Manager profiles.

11.2 Prompt Template Registry Definitions

Template 1: Candidate Application Summary Prompt (Summary Type)

***Object Binding:** Candidate_Application__c | ***API Name:** Candidate_Application_Summary_Prompt
***Context Instructions Matrix:**

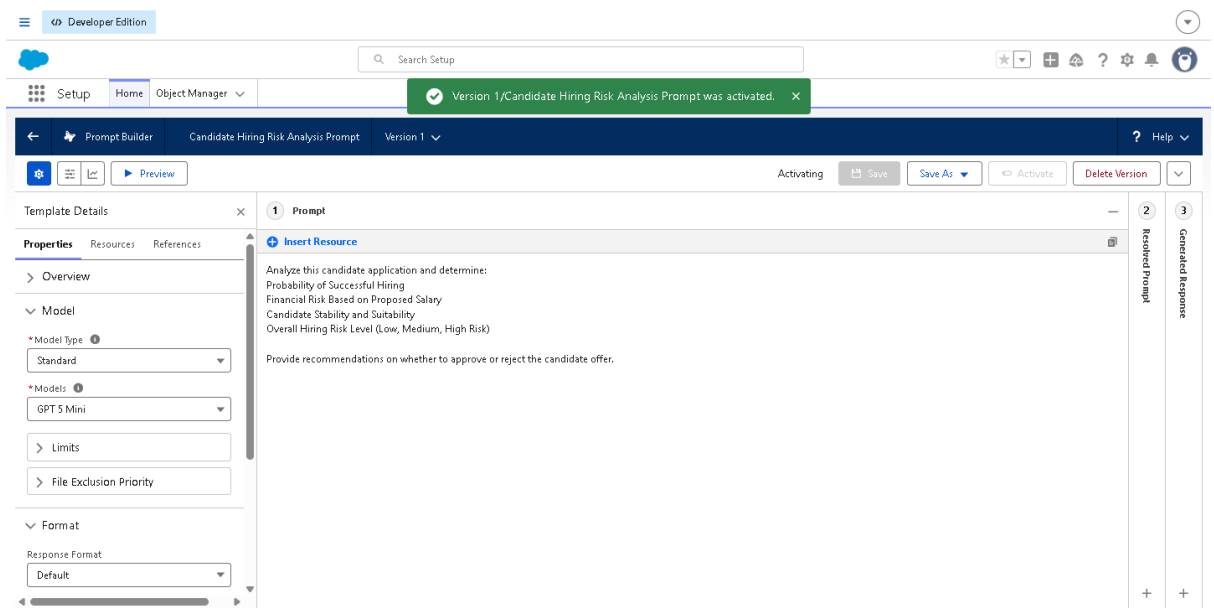
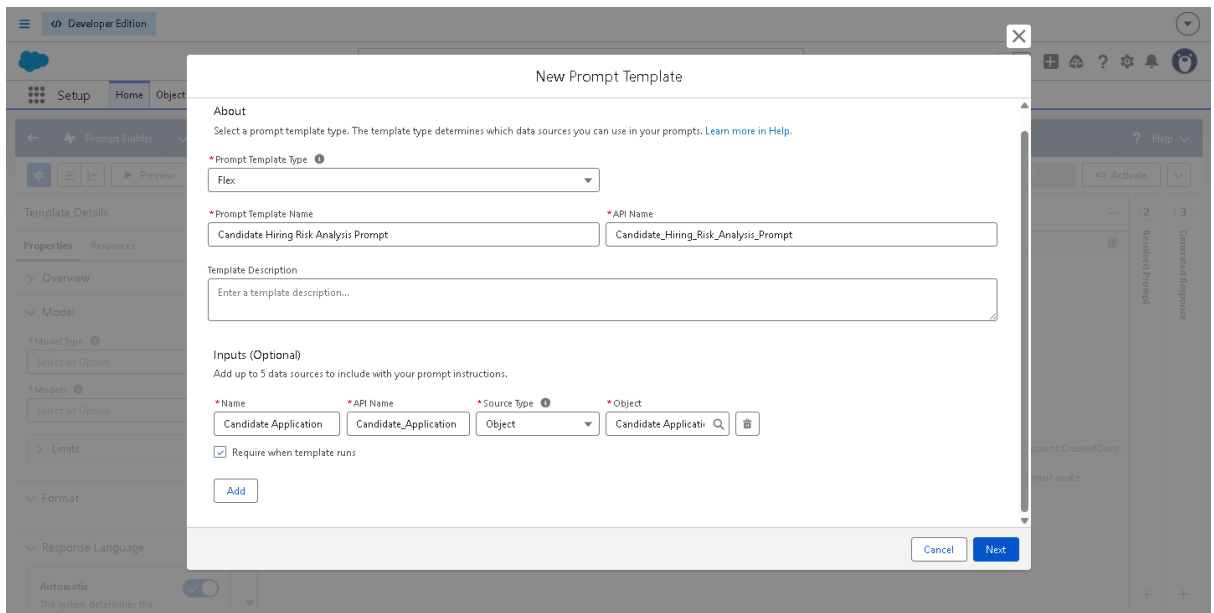
Summarize the candidate application in simple terms including: -
Candidate Name - Job Opening Applied - Experience Level -
Proposed Offer Amount - AI Candidate Score - Application Status
Highlight any risks, strengths, or special considerations that
may influence the hiring decision.



“Figure 11.1: Active GPT 5 Mini generative AI template Studio window for summary indexing.”

Template 2: Candidate Hiring Risk Analysis Prompt (Flex Type)
Object Binding: Candidate_Application__c | ***API Name:*** Candidate_Hiring_Risk_Analysis_Prompt
Context Instructions Matrix:

Analyze this candidate application and determine: 1. Probability of Successful Hiring 2. Financial Risk Based on Proposed Salary 3. Candidate Stability and Suitability 4. Overall Hiring Risk Level (Low, Medium, High Risk) Provide recommendations on whether to approve or reject the candidate offer.



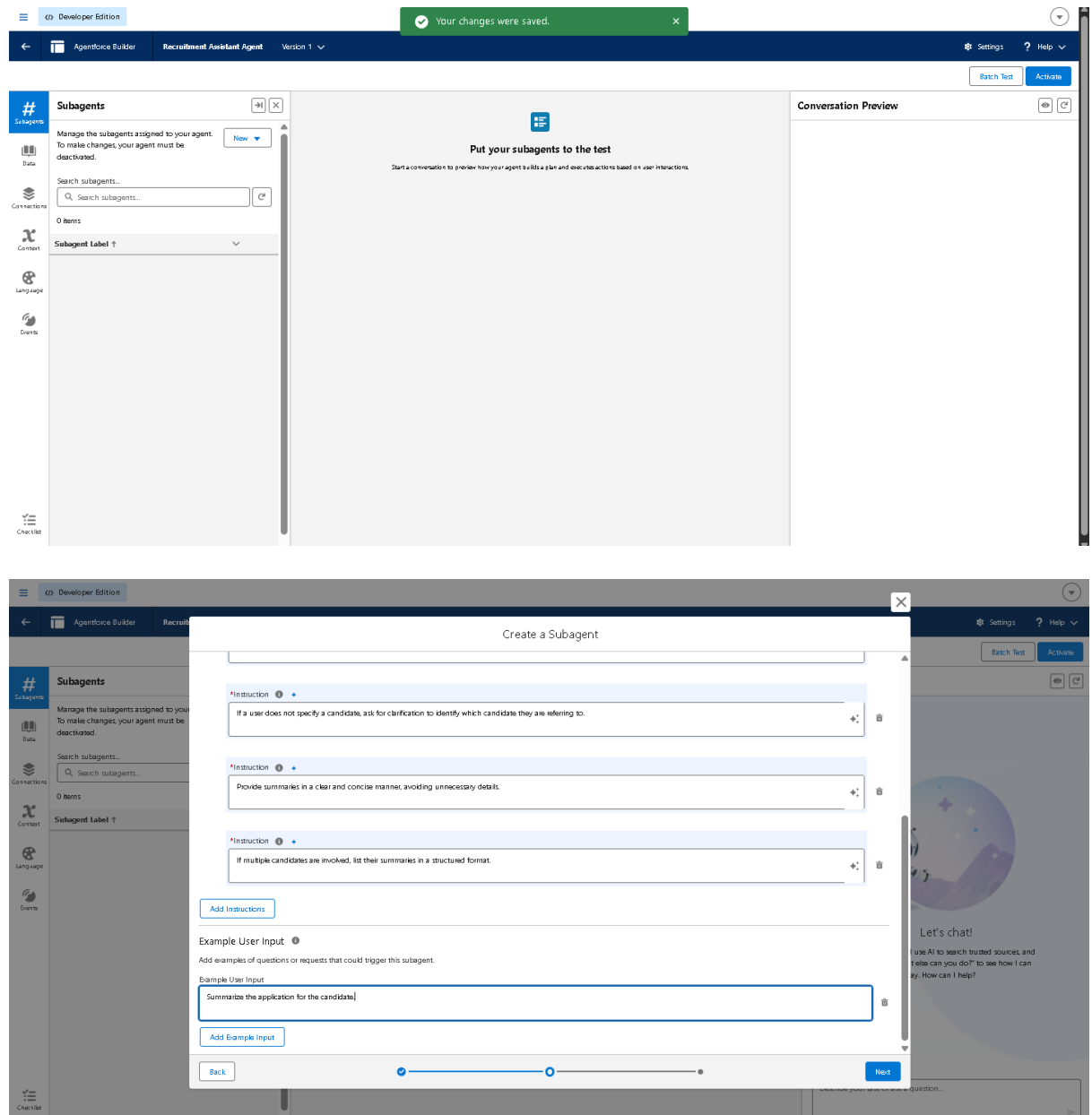
“Figure 11.2: Structuring object input bounds and activating the autonomous risk identification prompt matrix.”

11.3 Agent Deployment Architecture

The system packages these topics into an autonomous runtime engine labeled the **Recruitment Assistant Agent**. Inside the agent actions setup, the prompt templates are linked directly to operational targets: **Action: Generate Candidate Application Summary** calls Template 1, and **Action: Analyze Hiring Risk** calls Template 2.

The ****Agent Data Library**** grants the assistant secure, read-only access to Candidate (`Contact`), `Job\_Opening\_\_c`, and `Candidate\_Application\_\_c` data records. The

agent automatically respects native field-level security settings based on the running user profile, ensuring users only see information they are authorized to access.



“Figure 11.3: Deploying specialized, autonomous subagent instructions inside the core Agentforce Builder framework.”

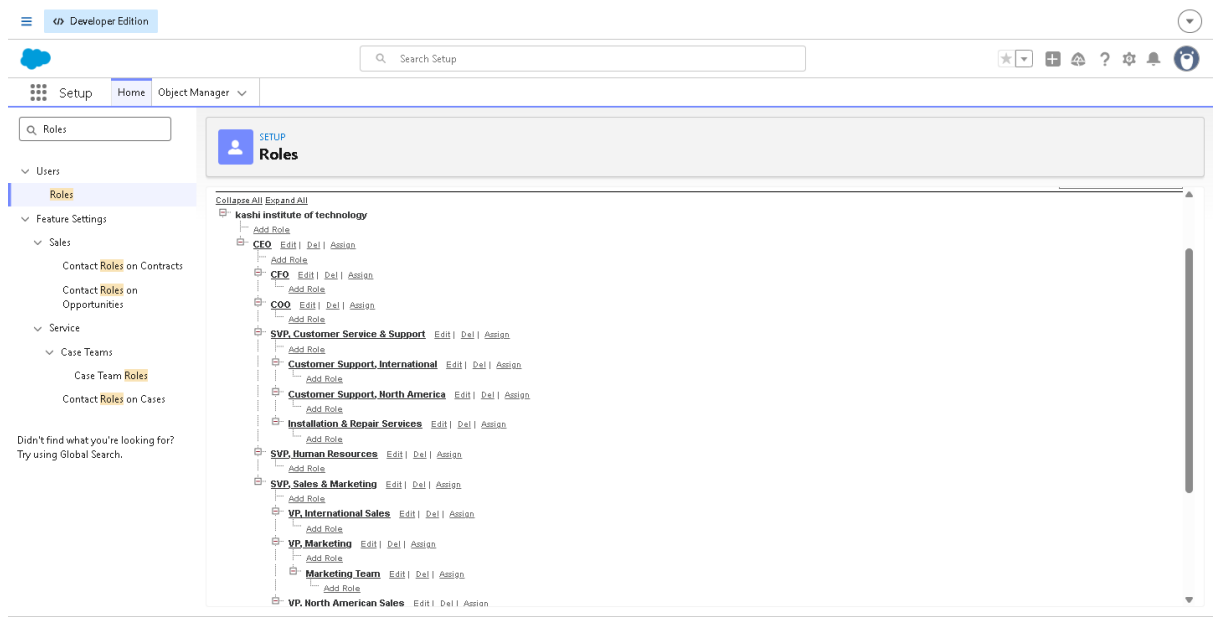
12. Security Architecture, Profiles, & Access Control

Data access boundaries are established using a structured role hierarchy combined with custom profile permission matrices, preventing unauthorized access across corporate tiers.

12.1 Organizational Visibility Hierarchy

The organization's hierarchy controls record visibility based on user roles:

Hiring Manager Role: Maintained directly beneath the CEO tier. This role grants full visibility across all underlying recruiter pipelines and candidate records. **Recruiter Role:** Maintained directly beneath the Hiring Manager role. This structure restricts visibility so frontline recruiters focus on their assigned openings and candidates.



“Figure 12.3: Mapping user access bounds via the active consolidated corporate Role Hierarchy tree directory.”

12.2 Profile Permission Security Configurations

Recruiter Profile Permissions Matrix

- **Custom App Default:** Automatically loads the `AI-Powered Recruitment & Candidate Management System` console application interface.
- **Object Permission Access Layer:**
 - Standard Contact (Candidates): Read, Create, Edit (Delete: Blocked)
 - Job Opening (Job_Opening__c): Read, Create, Edit (Delete: Blocked)
 - Candidate Application (Candidate_Application__c): Read, Create, Edit (Delete: Blocked)
 - Standard Reports and Dashboards: Read-Only Access
- **AI Capabilities Boundary:** Granted read-only visibility into the AI_Candidate_Score__c metric field. This profile is blocked from editing prompt frameworks or manually bypassing approval rules.

- **Session Policies:** Enforces automatic logouts after 2 hours of user inactivity.

Passwords expire every 90 days with a minimum length of 8 characters.

Figure 12.1: Administrative provisioning setup screen for ingesting a new Recruiter User into the platform.”

Hiring Manager Profile Permissions Matrix

- **Custom App Default:** Automatically loads the `AI-Powered Recruitment & Candidate Management System` console application interface.
- **Object Permission Access Layer:** Full Read, Create, Edit, and Delete permissions across all custom tracking objects, standard candidate contacts, reports, and team dashboard grids.
- **Policy Control Layer:** Full authority to approve or reject high-salary offers, modify prompt configurations, change model choices, and manually override AI evaluations.
- **Session Policies:** Enforces automatic logouts after 2 hours of user inactivity.

13. Data Migration & Legacy Ingestion Strategy

To replace legacy systems and spreadsheets cleanly, migration activities are split across specific tools to maintain relationship linkages:

46. **Candidate Data Migration Ingestion:** Historic candidate profiles are imported into the standard `Contact` table via the ****Data Import Wizard****, matching unique records against name and email fields.
47. **Job Openings & Applications Ingestion:** Bulk historical data loads are run through the ****Data Loader Engine**** tool to efficiently map data volumes into custom `Job_Opening__c` and `Candidate_Application__c` structures.

48. **Relational Mapping Validation:** Parent record IDs are verified during data mapping to ensure candidate applications reconnect correctly to their corresponding parent contact entries and job opening records.

14. System Verification & Quality Testing Logs

System configurations were verified using clear test case execution scripts inside a sandbox environment prior to final deployment:

ID	Test Scenario Description	Execution Input / Parameter Context	Expected Outcome Vector	Status
TC-01	Verify negative vacancy validation logic.	Set Vacancies__c = -3 on Job Opening record.	System blocks save; displays message: "Vacancies cannot be negative."	PASS
TC-02	Verify zero-value offer validation logic.	Set Offer_Amount__c = 0 on Candidate App record.	System blocks save; displays message: "Offer amount must be greater than zero."	PASS
TC-03	Verify out-of-bounds AI score limits.	Set AI_Candidate_Score__c = 120 on record save.	System blocks save; displays message: "AI Candidate score cannot exceed 100."	PASS
TC-04	Verify high-priority flow	Insert Candidate Application with an AI Score of 90.	Record-triggered flow executes	PASS

ID	Test Scenario Description	Execution Input / Parameter Context	Expected Outcome Vector	Status
	automation trigger.		immediately; fires email alert to owner.	
TC-05	Verify high-salary approval routing process.	Update Candidate Application with Offer = 1,500,000.	Flow intercepts save, locks application data, and routes to Manager.	PASS
TC-06	Verify data deduplication matching rule logic.	Insert Contact duplicate containing identical Email/Phone.	Duplicate rule blocks data ingestion and displays system alert text.	PASS
TC-07	Verify interactive creation screen flow.	Run Candidate_Application_Screen_Flow wizard layout.	Multi-screen inputs collect, process, and insert the record correctly.	PASS

15. Change Management, Support, & Platform Maintenance

To drive user adoption and maintain long-term system stability post-launch, specific change management steps are implemented:

- **Change Management & Adoption Strategy:** Agentforce chat interfaces and AI prompt summary generation blocks are embedded directly into standard Candidate record layout fields. This approach minimizes complexity by keeping insights inside the recruiter's daily workspace, requiring minimal user training.

- **System Monitoring Protocols:** System administrators review the ****Failed Flow Interviews**** log dashboard weekly to catch unexpected errors early. They also run periodic checks on prompt executions to evaluate accuracy, adjust instruction matrices, and prevent model drift.
- **Seasonal Update Management:** Testing protocols require running full regression evaluations across all automated flows and validation rules inside a sandbox environment before major Salesforce seasonal releases go live.

16. Strategic Platform Advantages & Future Scope

Realized Platform Advantages

- **Technical Debt Reduction:** Built completely with declarative tools, eliminating the development and maintenance costs of custom Apex code.
- **Pipeline Velocity Optimization:** Replaces disjointed tracking tools with automated record routing, reducing candidate drop-out rates.
- **Budget Compliance Control:** High-salary validation caps are enforced at the database level, preventing unauthorized compensation offers.
- **Accelerated Profile Screenings:** Grounded prompt templates summarize applicant records instantly, reducing manual evaluation time for hiring teams.

Documented Future Scope

- **Automated Interview Scheduling Integration:** Expand declarative flow paths to connect to external calendar applications for automated interview coordination.
- **Public Candidate Chat Portals:** Extend internal prompt capabilities into a public-facing Agentforce portal, allowing external candidates to track their application status directly.
- **Advanced Analytics Pipelines:** Build aggregate historical tracking dashboards to trace time-to-hire speeds and recruiter conversion metrics across different departments.

17. Conclusion & Reference Catalog

The successful deployment of the AI-Powered Job Recruitment & Candidate Management System demonstrates how organizations can use Salesforce's declarative framework to build robust enterprise applications. By connecting custom object relationships, automated validation steps, multi-tier approval paths, and Agentforce AI architectures, the system standardizes candidate tracking and protects data integrity without the overhead of custom programming.

Reference Catalog

59. *Salesforce Platform Security Controls & User Permission Configurations Manual* (help.salesforce.com)

60. *Salesforce Trailhead Frameworks: Declarative Automation Design & Flow Builder Best Practices* (trailhead.salesforce.com)
61. *Agentforce AI Implementation Guidelines & Prompt Engineering Design Architecture Standards* (developer.salesforce.com)
62. *SmartBridge Virtual Internship Program Track Documentation: Advanced Salesforce Architecture & Generative AI Enterprise Ingestion Frameworks.*